# Quality Information Document

## Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment Monitoring Service

OMI_CLIMATE_SL_GLOBAL_area_averaged_anomalies

OMI_CLIMATE_SL_GLOBAL_regional_trends

OMI_CLIMATE_SL_BALTIC_area_averaged_anomalies

OMI_CLIMATE_SL_BLKSEA_area_averaged_anomalies

OMI_CLIMATE_SL_EUROPE_area_averaged_anomalies

OMI_CLIMATE_SL_IBI_area_averaged_anomalies

OMI_CLIMATE_SL_MEDSEA_area_averaged_anomalies

OMI_CLIMATE_SL_NORTHWESTSHELF_area_averaged_anomalies

OMI_CLIMATE_SL_EUROPE_area_averaged_anomalies

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# CHANGE RECORD

| Issue | Date | § | Description of Change | Author | Validated By |
|---|---|---|---|---|---|
| 1.0 | 19/01/2018 | All | Creation | JF Legeais | |
| 2.0 | 26/07/2018 | All | New supporting material, new regional MSL. Use of the reprocessed DUACS DT-2018 products with updated temporal coverage | JF Legeais | |
| 2.1 | 25/01/2019 | All | Changes after review comments from Moi | JF Legeais | |
| 2.2 | 16/01/2020 | | Addition of global mean sea level corrected for TOPEX-A drift | JF Legeais | |
| 2.3 | 25/03/2020 | Ref | Update of reference | JF Legeais | |
| 2.4 | 20/08/2020 | | Regional uncertainties and reference update | JF Legeais | |
| 3.0 | 10/09/2021 | | Update after full OMI reprocessing | G Taburet | M-I Pujol |
| 4.0 | 05/05/2022 | All | Use new template<br>revised after MOi review | G.Taburet | M-I Pujol |
| 4.1 | 05/04/2023 | II<br>III | Global uncertainties and reference update | P Veillard | M-I Pujol |
| 5.0 | 15/06/2023 | | Use "myinteruim" datasets<br>Nomenclature change | M-I Pujol | M-I Pujol |
| 6.0 | 001/02/2024 | All | Applying global TPA correction and updated regional GIA mean to regional timeseries.<br>Applying global TPA correction and regional GIA to map of trends.<br>Adding global uncertainty envelope timeseries.<br>Adding Europe region. | P. Veillard | MI. Pujol |

| | | | | | |
|---|---|---|---|---|---|
| 7.0 | 31/01/2025 | All | Update with new v2024 C3S products<br>No global TPA correction applied (no longer consistent with new v2024 products)<br>New temporary reference period for the computation of the GMSL parameters (trend and acceleration) | V. Quet | K Grassi |

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# I INTRODUCTION

## I.1 Product name

OMI_CLIMATE_SL_GLOBAL_area_averaged_anomalies
OMI_CLIMATE_SL_GLOBAL_regional_trends
OMI_CLIMATE_SL_BALTIC_area_averaged_anomalies
OMI_CLIMATE_SL_BLKSEA_area_averaged_anomalies
OMI_CLIMATE_SL_EUROPE_area_averaged_anomalies
OMI_CLIMATE_SL_IBI_area_averaged_anomalies
OMI_CLIMATE_SL_MEDSEA_area_averaged_anomalies
OMI_CLIMATE_SL_NORTHWESTSHELF_area_averaged_anomalies

## I.2 Dataset:

omi_climate_sl_global_area_averaged_anomalies
omi_climate_sl_global_regional_trends
omi_climate_sl_baltic_area_averaged_anomalies
omi_climate_sl_blksea_area_averaged_anomalies
omi_climate_sl_europe_area_averaged_anomalies
omi_climate_sl_ibi_area_averaged_anomalies
omi_climate_sl_medsea_area_averaged_anomalies
omi_climate_sl_northwestshelf_area_averaged_anomalies

## I.3 Indicator evaluation method

The CMEMS sea level OMIs are derived from the DUACS delayed-time two-satellite merged altimeter sea level gridded products, in DT-2024 version, distributed by the Copernicus Climate Change Service (C3S) and also available in the CMEMS catalogue (product SEALEVEL_GLO_PHY_CLIMATE_L4_MY_008_057, "my" dataset when available, "myint" after). Dedicated documentation for these products is available on C3S page:

https://cds.climate.copernicus.eu/cdsapp#!/dataset/satellite-sea-level-global?tab=doc

The C3S merged products are specifically processed focusing on the homogeneity and stability of the sea level record. They are thus dedicated to the monitoring of the long-term evolution of the sea level for climate applications and the analysis of Ocean/Climate indicators. Compared with the CMEMS delayed time merged gridded sea level products, the main differences are:

The number of altimeters used in the satellite constellation: all available altimeters are included in the CMEMS products whereas a steady number (two) of altimeters are included in the C3S products. Previous studies (Dibarboure et al., 2014; Pascual et al., 2006) underscored the necessity of a minimum of a two-satellite constellation for the retrieval of mesoscale signals. Within the production process, the long-term stability and large-scale changes are built upon the records from the reference missions (TOPEX-Poseidon, Jason-1, Jason-2,Jason-3 and Sentinel-6A) used in both CMEMS and C3S products. The additional missions (e.g., up to 7 additional missions in 2019) are homogenized with respect to the reference missions and contribute to improve the sampling of mesoscale processes, provide the high-latitude coverage and increase the product accuracy. However, the total number of satellites strongly varies during the altimetry era and some biases may appear with the introduction of a new satellite flying on a drifting orbit, which may affect the stability of the global and regional MSL. Even if the spatial sampling is reduced with less satellites, the risk of introducing such anomaly is thus reduced in the C3S products and the stability is improved. In the CMEMS products, the stability is ensured by the reference missions and the mesoscale errors are reduced due to the improved ocean surface sampling thanks to the use of all satellites available in the constellation.

The reference used to compute the sea level anomalies is a mean sea surface (MSS) for all missions in the C3S products whereas a mean profile of sea surface heights is used along the theoretical track of the satellites with a repetitive orbit in the CMEMS products. These mean profiles increase the local accuracy of the sea level estimates, but the combined use of both MSS and mean profiles for successive missions in the CMEMS merged products can be at the origin of some biases affecting the sea level stability. Even if this has a minor impact at global scale, the stability of the regional MSL can be affected (particularly in the Mediterranean Sea and Black Sea). So, the systematic use of the MSS for all missions contributes to ensure the MSL stability in the C3S products and the accuracy of the CMEMS products is increased with the use of the mean profiles for repetitive missions.

The temporal and spatial scales used in the mapping process (Optimal Interpolation) have been reviewed specifically for the CMEMS production and are now different in C3S and CMEMS sea level products. The spatial and temporal correlation scales have been adjusted (using an ensemble approach)

to take into account the larger number of altimeters used in the CMEMS configuration and to allow a better resolution of the mesoscale signal. Adjusting theses scales yields significantly better results in areas of high ocean variability, especially at scales between 65 and 500km.

For these reasons, the C3S products are used to produce the CMEMS ocean monitoring indicators.

The altimeter satellite multi-mission gridded sea surface heights and derived variables are computed with respect to a twenty-year mean reference period (1993-2012). Up-to-date altimeter standards are used to estimate the sea level anomalies (Lievin et al., 2020). More details on the altimeter and processing algorithms are available in (Pujol et al., 2016).

Between 1993 and 1998, satellite altimetry sea level is known to have been affected by a TOPEX-A instrumental drift (WCRP Global Sea Level Budget Group, 2018 and Legeais et al., 2020 in OSR4(Von Schuckmann et al., 2020). This anomaly has been characterised in different studies (see e.g., (Valladeau et al., 2012; Watson et al., 2015; Dieng et al., 2017; Beckley et al., 2017). However, the reprocessing of Topex data in GDR-F standards lead to major changes when compared with MGDR standards (used until v2021 C3S products) and previous studies are no longer updated to describe the probable drift in Topex A data. On-going work aim at deliver a new empirical correction for this period but it will not make it to the delivery of this version of OMI products.

Contrary to near real-time (NRT) sea level products, the stability and accuracy of the delayed-time products make them adapted to climate applications and ocean monitoring indicators. The timeliness of the sea level products reaches about 5 months due to the timeliness of the input data, the centered processing temporal window and the validation process. It is not recommended to use the NRT products for climate studies, especially for the completion of the MSL time series close to present time. Indeed, some biases can be found at the DT/NRT transition, especially for the computation of area-averaged MSL. Works are ongoing to combine the DT/NRT products and produce near real-time ocean monitoring indicators.

**Dataset**: omi_climate_sl_global_area_averaged_anomalies

Time series of global area averaged sea level maps from satellite altimetry. The time series is corrected from global GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered.

**Dataset**: omi_climate_sl_global_regional_trends

Map of sea level trends from sea level maps from satellite altimetry. At each grid point, the trend is estimated on the time series corrected from regional GIA correction and adjusted from annual and semi-annual signals.

**Dataset**: omi_climate_sl_baltic_area_averaged_anomalies

Time series of Baltic Sea area averaged sea level maps from satellite altimetry. The time series is corrected from regional mean GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered.

**Dataset**: omi_climate_sl_blksea_area_averaged_anomalies

Time series of Black Sea area averaged sea level maps from satellite altimetry. The time series is corrected from regional mean GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered.

**Dataset**: omi_climate_sl_ibi_area_averaged_anomalies

Time series of Iberian-Biscay-Irish Sea area averaged sea level maps from satellite altimetry. The time series is corrected from regional mean GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered.

**Dataset**: omi_climate_sl_medsea_area_averaged_anomalies

Time series of Mediterranean Sea area averaged sea level maps from satellite altimetry. The time series is corrected from regional mean GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered.

**Dataset**: omi_climate_sl_northwestshelf_area_averaged_anomalies

Time series of North-West Shelf Sea area averaged sea level maps from satellite altimetry. The time series is corrected from regional mean GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered.

**Dataset**: omi_climate_sl_europe_area_averaged_anomalies

Time series of Northeast Atlantic and adjacent regional seas area averaged sea level maps from satellite altimetry. The time series is corrected from regional mean GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered.

## I.4 CMEMS products used as input for the computations

| Product name (and Type) | Documentation |
|---|---|
| SEALEVEL_GLO_PHY_CLIMATE_L4_MY_008_057:<br>Gridded product merging measurements from 2 altimeter missions.<br>'my' dataset is used when available; "myint" dataset is used after. | QUID and PUM available on the product documentation page :<br>https://resources.marine.copernicus.eu/product-detail/SEALEVEL_GLO_PHY_CLIMATE_L4_MY_008_057/DOCUMENTATION |

# II SCIENTIFIC VALIDATION

Scientific validation results and discussion of the uncertainties can be found in the second Copernicus Marine Service Ocean State Report (Dieng et al., 2017), in (Guérou et al., 2023) and in (Quet et al., 2024)[in prep.]: the main sources of errors are related to several altimeter standards (wet troposphere correction and orbit solution), the instabilities of the altimeter parameters and the estimation of inter-mission biases.

Dieng et al., 2017 show that the ensemble mean of the global MSLs produced by different international altimeter groups is very similar to the sum of the components that contribute to the sea level for the period 1993-2015, suggesting the closure of the sea level budget. However, non-closure of Global Mean Sea Level Budget since 2016 has been identified with part of it coming from salinity issues of Argo measurements (Barnoud et al., 2021).

For the trend estimation of the mean sea level in the different CMEMS regions, only errors related to the altimeter observation system have been considered. Errors related to the sea level internal variability should be also considered additionally. Indeed, the local variability is generated by regional changes in winds, surface atmospheric pressure and ocean currents which averaged out at the global scale (Stammer et al., 2013) but this can significantly contribute to the MSL uncertainty at the basin scale.

Additional information regarding the validation of the sea level products and the description of the associated errors and uncertainties are available in (Ablain et al., 2015; Legeais et al., 2018; SL_cci_CECR, 2016; Taburet et al., 2019; Guérou et al., 2023; Quet et al., 2024)[in prep.]). In these papers, the errors and associated uncertainties have been described distinguishing the mean sea level at different spatial (global and regional) and temporal (long-term trend, inter-annual signals and annual signal) scales.

Regarding the global mean sea level evolution, (Ablain et al., 2019; Guérou et al., 2023; Quet et al., 2024) have focused on the estimation of errors affecting the global mean sea level record in a budget approach where individual error contributions are combined and used to estimate the total error variance-covariance matrix (available for the users). Then, this matrix is used to estimate the uncertainty on the trend (and acceleration) of the global mean sea level record during any period extracted from the altimetry era. Resulting uncertainties depending on the period considered are summarized on Figure 1.

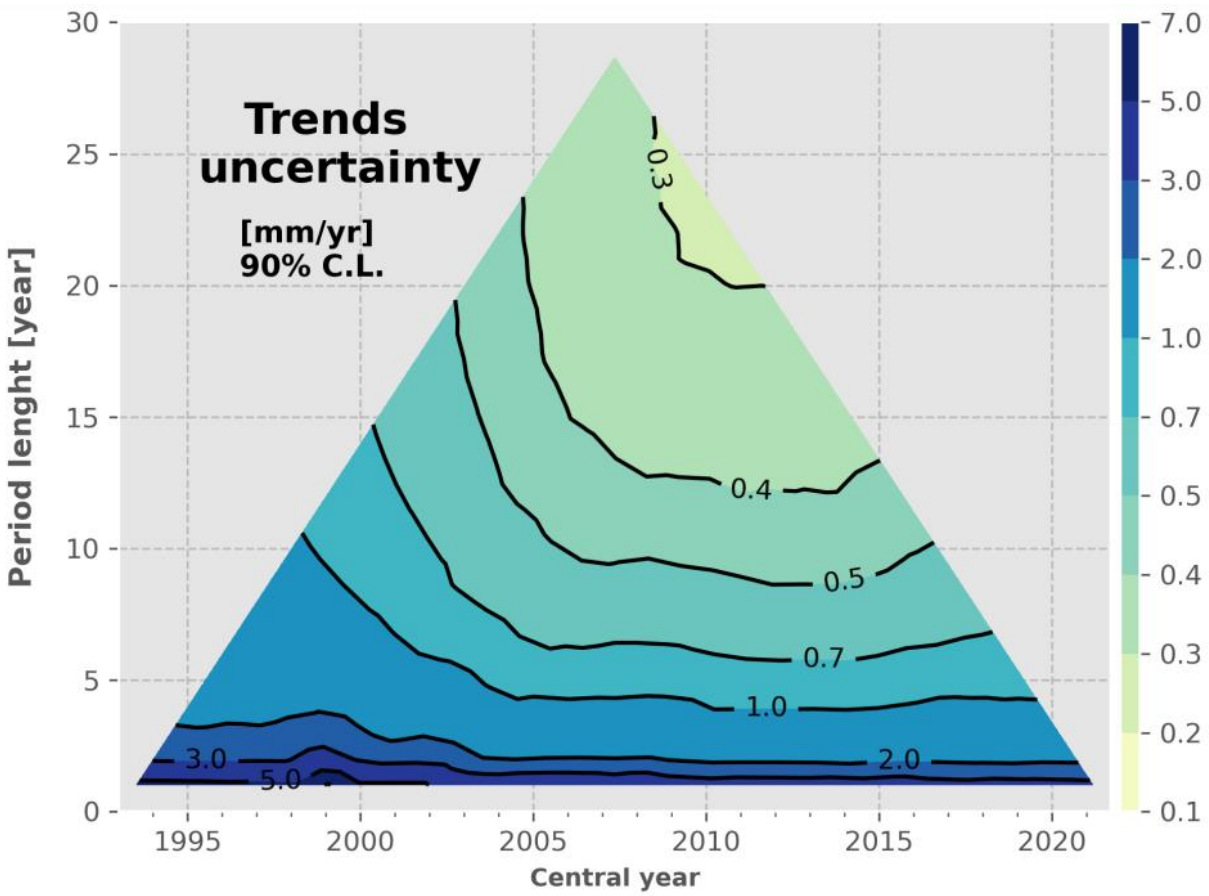


*Figure 1: Global mean sea level trend uncertainty in mm/yr at the 90% confidence level as a function of the time series length and period. The y-axis represents the length of the record from 2 to 30 years and the x-axis represents the central date of the sub-record used. Credits: (Guérou et al., 2023).*

Over the full record, the global mean sea level uncertainty on the trend is of ± 0.3 mm/yr and the uncertainty on the acceleration is of 0.05mm/yr² are estimated. Note that uncertainties below 0.3 mm/yr are reached over 20 years periods centered around 2010, which exclude the higher error associated with the TOPEX-A record.

The regional sea level trend uncertainty has an average estimate of 0.83 mm/yr with local values ranging from 0.78 to 1.22 mm/yr depending on the regions. These values are only related to the errors of the altimeter instrumental observing system (WCRP Global Sea Level Budget Group, 2018).

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